Morgan Scalabrino

Mathematics and Artificial Intelligence Master 2 Student
Université Paris Saclay

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About me - Morgan Scalabrino

My project:

Improve my fundamental and practical knowledge in Computer Vision, Multimodal Models and Self Supervised Learning applied to Health and Biology to continue with a PhD in these fields.

Experiences:

Internship Télécom Paris, IMAGES team — **Summer 2026 – 6 months**
Supervised by Maxime Di Folco — *Paris Saclay*
Implementing hierarchical alignment methods to build Vision Language Models for Chest X Ray zero shot classification and retrieval. Implementation of SAIL paper in a MedCLIP setting with hierarchical regularization. Designing new hierarchical alignment contrastive objective with multilabel capacity.

Internship INRIA, THOTH team — **Summer 2025 – 5 months**
Supervised by Hadrien Hendrikx — *Grenoble*
Using State of the art detection and classification deep learning and computer vision methods on soil scanners (DINO v2, ConvNeXt v2, MAE). Implementation of hierarchical classification, Masked autoencoder self-supervised learning and custom architectures.

Internship INRIA, BIOCORE team — **Summer 2023 – 2 months**
Supervised by Olivier BERNARD — *Sophia Antipolis*
Mathematical modeling in Matlab of microbial ecosystems integrating feedforward neural networks into a mechanistic model of ordinary differential equations.

Internship Institut de Biologie de Valrose — **January 2023**
Supervised by Franck Delaunay — *Nice*
Introduction to research in biology through the different team of the institute

Education:

Master 1 & 2 Mathematics and Artificial Intelligence — **2024-2026**
Université Paris Saclay: course in common with MVA's ENS Paris Saclay program and Centrale Supelec Engineering School

Mathematics: Optimization, Mathematical Foundation of Deep Learning and Generative Models, Methods in Regression and Classification, Decision Theory and Convergence Properties of seen models.

Machine Learning: Object Recognition, Representation Learning, Optimization for Computer Vision, Natural Language Processing, Reinforcement Learning, Generative Models, Geometric Processing and Geometric Deep Learning, Clustering, Dimension Reduction and Visualization.

Master Projects:
- Project Management with Git, presentation with Streamlit.
- Presentation and Implementation of research paper: Multimodal Deep Learning with Feature Wise transformation, Missing Data EM Algorithm, Self Supervised Learning of Reliable Point Representation, CLIP.
- 5 Data Challenges on regression and classification for time series and tabular data.

Double Bachelor in Mathematics and Biology — **2021-2024**
Université Côte d'Azur, Nice
Mathematics & Biology & IT: Foundations in math and biology. Skills in Python, Linux, Scilab, R, algorithmic.

Bachelor Projects: Construction of agent-based model (Netlogo) and of a tool with graphic interface which analyzes genomic databases (Python).

Languages:

English	Level C1 (TOIEC 975/990)
Spanish	Level B1

Hobbies:

-Solo unsupported hiking (1 week/month)
-Whitewater kayaking
-Biking

Programming Languages/tools:

Python: Pytorch, Transformers, OpenCLIP, FAISS, lightly, lightning, Scikit-learn, Numpy, Panda, Matplotlib

And other tools/languages: **GIT**, **linux** system gestion, Zotero, Nvidia NSight, R, Matlab/Scilab, SQL.